



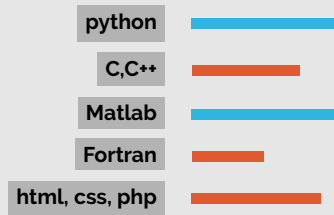
About me

Researcher interested in the intersection of Physics and AI. Data-Driven Physics Discovery and Physics-Inspired ML theory

Areas of specialization

- Physics • Machine Learning • Geophysics • Seismic Inversion • High Energy Physics

Programming



Other Skills

- Proficient Arabic Speaker • Freelance Photographer

Interests

Physics / AI / Photography
Physics / AI / Photography
Physics / AI / Photography

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in ibrahimelsharkawy

WORK EXPERIENCE

March 2025 -	Doctoral Researcher LAWRENCE BERKELEY NATIONAL LABORATORY · Berkeley, CA • Researcher at the National Energy Research Scientific Computing Center (NERSC) building Foundation Models for Scientific Point Clouds/Graphs for High Energy Physics, Cosmology and Molecular Dynamics.	
2024-2025	Research Geophysicist Intern ExxonMobil · Houston, TX • Researched and developed pioneering machine learning parameter estimation methods for 3D-seismic inversion with the potential of enabling the inversion of traditionally degenerate earth properties • Researched and developed novel ML anomaly detection tools for 4D Seismic.	
2018-2023	Research Geophysics and Software Engineering Intern PETROLEUM GEOSERVICES · Houston, TX • Researched ML methods for first break picking in seismic. • Built a seismic ML anomaly detector • Worked on cloud migration and microservices	

EDUCATION

2025-	Physics PHD · University of Toronto	
2023-2025	Physics MSc · University of Illinois Urbana-Champaign	
2019-2023	Computational Physics B.S. · Rice	
2019-2023	Applied Mathematics and Philosophy B.A. · Rice	

AWARDS AND HONORS

2025	University of Toronto Con-naught International Fellow
2024	1st Place Higgs Uncertainty Challenge @ NeurIPS, CERN
2023	3-Year UIUC Graduate College Fellowship
2021	J & M Graham Physics Scholar
2019	Shell Eco-Marathon
2016	Eagle Scout

RESEARCH EXPERIENCE

2025-	Foundation Models For Science LBNL · PROF. WAHID BHIMJI, PROF. BEN NACHMAN · Berkeley, CA • Building foundation models for scientific point clouds with cross-domain transfer . • Researching optimal pre-training objectives for foundation models in science, where simulators can produce <i>labeled</i> data at scale.	
2023-2025	Neural Uncertainty Quantification and Parameter Estimation UIUC · PROF. YONI KHAN PROF. BEN HOOBERMAN · Urbana, IL • Used an effective field theory of NN to successfully predict initialization-driven ensemble uncertainty without training [1]. • Built a competition-winning ML methodology robust to systematic sources of error [2]	
2019-2023	CERN CMS Higgs-W-W-Boson Analysis RICE PHYSICS · PROF. KARL ECKLUND · Houston, TX • Working on the CMS Same-Flavor W^+W^- Boson Pair Cross Section Measurement in Proton-Proton Collisions at $\sqrt{s} = 13\text{TeV}$.	

SELECT PAPERS

1. Elsharkawy, Ibrahim, et al. "Uncertainty Quantification from Ensemble Variance Scaling Laws in Deep Neural Networks." Machine Learning: Science and Technology, vol. 6, no. 3, 26 Aug. 2025, p. 035040

2. Elsharkawy, Ibrahim, and Yonatan Kahn. "Contrastive Normalizing Flows for Uncertainty-Aware Parameter Estimation." arxiv.org/abs/2505.08709.

3. Elsharkawy, Ibrahim, et al. "OmniMol: Transferring Particle Physics Knowledge to Molecular Dynamics with Point-Edge Transformers." ArXiv.org, 2026, arxiv.org/abs/2601.10791.

4. Wahid Bhimji, et al. "FAIR Universe HiggsML Uncertainty Dataset and Competition." , In Proceedings, NeurIPS 2025 Dataset and Competition Track

5. Mikuni, Vinicius, et al. "OmniCosmos: Transferring Particle Physics Knowledge across the Cosmos." ArXiv.org, 2025, arxiv.org/abs/2512.24422.

SELECT TALKS/POSTERS

- 1. (Invited Talk)** *1st Place Competition Milestone (Ensembles and Uncertainty Quantification)* in The Challenge of Handling Uncertainties in Fundamental Science @ NeurIPS 2024
- 2. (Invited Talk)** *Contrastive Normalizing Flows for Uncertainty Aware Parameter Estimation* @ CERN 7th Inter-Experimental LHC Machine Learning Workshop 2025
- 3. (Poster)** Wahid Bhimji, et al. *"FAIR Universe HiggsML Uncertainty Dataset and Competition."*, NeurIPS 2025 Dataset and Competition Track
- 4. (Poster)** I. Elsharkawy, et.al, *Uncertainty Quantification from Scaling Laws in Deep Neural Networks*, in Machine Learning and the Physical Sciences Workshop @ NeurIPS 2024
- 5.(Poster)** I. Elsharkawy, et.al,*Pre-Training For Science: A study on Foundation Model Training Objectives*, Stanford Center of Decoding the Universe Forum 2025

TEACHING EXPERIENCE

- 1. TA and Lecturer** *University of Toronto, 2026, Physics 2108/2109, "The Physics of Machine Learning"*
- 2. TA** *University of Toronto, 2025, Physics 252, "Thermal Physics"*
- 3. TA** *University of Illinois Urbana-Champaign, 2024, Physics 413 "Optics Lab"*

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